Problem 6

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Problem 1 (a) *Find the following definite integral:*

$$\int_{-4}^{4} \frac{x^2 \sin^3(x)}{\sqrt{x^4+1}}\, dx$$

Explanation. Let $f(x) = \dfrac{x^2 \sin^3(x)}{\sqrt{x^4+1}}$. Then notice that

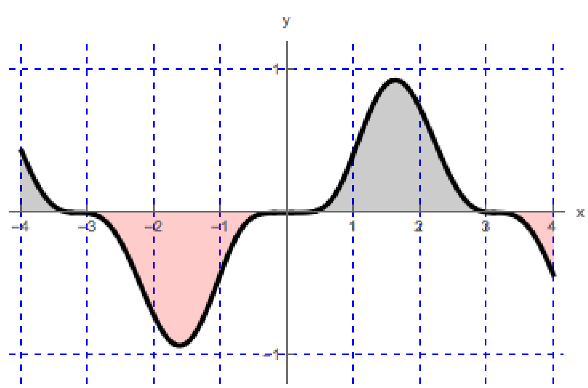
$$\begin{aligned} f(-x) &= \frac{(-x)^2 \sin^3(-x)}{\sqrt{(-x)^4+1}} \\ &= \frac{x^2(-\sin(x))^3}{\sqrt{x^4+1}} \quad \text{(since } \sin(x) \text{ is an odd function)} \\ &= \frac{-x^2 \sin^3(x)}{\sqrt{x^4+1}} \\ &= -f(x). \end{aligned}$$

Thus, f is an odd function and therefore

$$\int_{-4}^{4} \frac{x^2 \sin^3(x)}{\sqrt{x^4+1}}\, dx = 0.$$

We can illustrate our computation with the graph of the function f, where

$f(x) = \dfrac{x^2 \sin^3(x)}{\sqrt{x^4+1}}$, for $-4 \le x \le 4$.



(b) *Suppose that f is an even function. Given that* $\int_0^6 f(x)\,dx = 13$, *find* $\int_{-6}^{6} (5f(x) + 14)\,dx$.

Explanation. First notice that

$$\int_{-6}^{6} (5f(x) + 14)\,dx = 5\int_{-6}^{6} f(x)\,dx + \int_{-6}^{6} 14\,dx. \tag{1}$$

Since f is even, we know that

$$\int_{-6}^{6} f(x)\,dx = 2\int_{0}^{6} f(x)\,dx = 2(13) = 26. \tag{2}$$

We also know that $g(x) = 14$ is also an even function. We have: $\int_{-6}^{6} 14\,dx =$ $2\int_0^6 14\,dx$

$$2\int_0^6 14\,dx = 2(14)(6) = 168. \tag{3}$$

Then substituting equations (2) and (3) into equation (1) gives:

$$\int_{-6}^{6} (5f(x) + 14)\,dx = 5(26) + 168 = 130 + 168 = 298.$$